

SQL CITY HOMICIDE CASE FILE

A SQL-Driven Crime Investigation — Technical Project Report

CASE NO. 20180115-MRD-SQLCITY

Status: CLOSED — Shooter Convicted, Mastermind Identified

Prepared using the SQL Murder Mystery dataset
Analytical Method: Structured Query Language (SQL)

Date of Report: September 2026

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1. Introduction

On the night of January 15, 2018, a homicide was reported in SQL City. What began as a single row in a `crime_scene_report` table unraveled into a multi-layered investigation spanning witness interviews, gym membership records, vehicle registrations, and financial motive. This report documents the complete analytical process used to solve the case using nothing but a relational database and Structured Query Language (SQL).

The purpose of this project is twofold: first, to demonstrate practical SQL querying skills — filtering, joins, pattern matching, table creation, and multi-table reasoning — applied to a realistic, narrative-driven dataset; and second, to present those findings as a professional investigative report, complete with visual aids, insights, and recommendations for future data-driven casework.

The dataset used is the popular "SQL Murder Mystery" schema, which simulates a city's law-enforcement and civic records: crime scene reports, persons, interviews, gym memberships and check-ins, driver's licenses, and (implicitly) financial and event records referenced in witness testimony. Every query in this report is reproduced from the original analysis file and annotated with investigative context.

2. Story of Data

Every dataset tells a story if you ask it the right questions. This one tells the story of a killer-for-hire, a wealthy mastermind, and the data trail that connected them.

2.1 The Crime

January 15, 2018 — SQL City. A gunshot breaks the evening quiet. By the time officers arrive, the perpetrator is gone — but two people saw enough to matter.

The case opens with a single lookup: a murder recorded in SQL City on January 15, 2018. Security footage places two witnesses at the scene. Investigators trace the first witness to the last house on Northwestern Drive, and the second — a woman named Annabel — somewhere along Franklin Avenue.

2.2 The Witnesses Speak

The two witnesses are identified as Morty Schapiro (last house on Northwestern Dr.) and Annabel Miller (103 Franklin Ave.). Their interview transcripts become the backbone of the entire investigation:

Morty: "I heard a gunshot and then saw a man run out. He had a 'Get Fit Now Gym' bag. The membership number on the bag started with '48Z'. Only gold members have those bags. The man got into a car with a plate that included 'H42W'."

Annabel: "I saw the murder happen, and I recognized the killer from my gym when I was working out last week on January the 9th."

Two independent leads — a partial gym membership number and a partial license plate — combined with a specific gym check-in date, gave investigators three converging filters instead of one.

2.3 Following the Gym Bag

A search of gold-tier gym members with a membership ID beginning in "48Z" returned two candidates: Joe Germuska and Jeremy Bowers. Cross-referencing this pair against everyone who checked into Get Fit Now Gym on January 9 — the day Annabel says she recognized her future killer — confirmed that both men had, in fact, visited the gym that day, so a second, independent identifier was needed to break the tie.

2.4 Following the License Plate

Turning to Morty's partial plate — "H42W" — investigators searched the driver's license table for any plate containing that fragment, regardless of position. Three plates matched. Joining the driver's license table to the person table by license ID revealed the drivers' identities, and only one of them lined up with the gym-suspect shortlist: Jeremy Bowers.

2.5 The Shooter Confesses — and Names His Employer

Jeremy Bowers' own interview closed the loop on the shooting itself, but opened a bigger question: he was not acting alone.

Jeremy: "I was hired by a woman with a lot of money. I don't know her name but I know she's around 5'5" (65") or 5'7" (67"). She has red hair and she drives a Tesla Model S. I know that she attended the SQL Symphony Concert 3 times in December 2017."

This testimony reframed the entire case. Jeremy Bowers was the shooter, but the story of the data did not end with him — it pointed toward a wealthy mastermind hiding behind a physical description, a taste for luxury cars, and a pattern of concert attendance.

3. Data Splitting and Preprocessing

Unlike a machine-learning workflow, this investigation did not split data into training and test sets. Instead, "preprocessing" took the form of progressively filtering and re-joining relational tables so that each step consumed only the subset of data relevant to the current lead. The core tables involved were:

Table	Role in Investigation
crime_scene_report	Anchor record; identifies the date, city, and type of crime.
person	Master directory of city residents, used to resolve names and addresses.
interview	Free-text witness and suspect statements, keyed by person_id.
get_fit_now_member	Gym membership records, including tier (gold/silver) and ID prefix.
get_fit_now_check_in	Gym attendance log, used to confirm presence on a specific date.

drivers_license	Vehicle and physical-description data (plate, height, hair color, car make/model).
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Two derived tables were created mid-investigation to persist intermediate results and simplify later filtering:

- suspects — gold members whose ID begins with the partial membership number from Morty's testimony.
- full_data — a left join of drivers_license and person on license ID, consolidating physical description, vehicle details, and identity into a single queryable view.

This mirrors good analytical practice in any SQL project: materializing intermediate, reusable subsets (via CREATE TABLE AS) rather than repeating complex joins in every subsequent query.

4. Pre-Analysis

Before any suspect narrowing could occur, the investigation had to establish the basic facts of the case and locate the two witnesses. This phase relied entirely on simple SELECT and WHERE filtering — no joins, no aggregation.

```
SELECT * FROM crime_scene_report
WHERE date = 20180115 AND city = 'SQL City' AND crime_type = 'murder';
```

Locating the witnesses required pattern-based filtering and sorting rather than an exact match, since only partial address and name clues were available:

```
SELECT * FROM person
WHERE address_street_name = 'Northwestern Dr'
ORDER BY address_number DESC; -- last house on the street

SELECT * FROM person
WHERE name LIKE 'Annabel%' AND address_street_name = 'Franklin Ave';
```

This pre-analysis stage established the two named witnesses — Morty Schapiro and Annabel Miller — whose testimony would drive every subsequent query.

5. In-Analysis

The in-analysis phase is where the investigation moved from single-table lookups to genuine multi-table reasoning, narrowing an entire city's population down to a single named suspect.

5.1 Retrieving Testimony

```
SELECT * FROM interview WHERE person_id IN (14887, 16371);
```

5.2 Filtering Suspects by Membership Clue

Morty's partial gym membership number and gold-tier detail were used to materialize a suspects table:

```
CREATE TABLE suspects AS
(SELECT * FROM get_fit_now_member
 WHERE id LIKE '48%' AND membership_status = 'gold');
```

This returned two candidates — Joe Germuska and Jeremy Bowers — both gold members with IDs beginning in the right range.

5.3 Cross-Referencing Gym Check-ins

```
SELECT * FROM get_fit_now_check_in
WHERE check_in_date = 20170109
AND membership_id IN ('4D5R1', '4KB72');
```

Both suspects had checked in on the relevant date, so this filter alone could not break the tie — a reminder that in real investigative data work, not every query yields a clean, singular answer. A second independent variable was required.

5.4 Cross-Referencing the License Plate

```
SELECT * FROM drivers_license
WHERE plate_number LIKE 'H42W%'
OR plate_number LIKE '%H42W%'
OR plate_number LIKE '%H42W';
```

Three plates matched the partial fragment, but none of the raw results included a suspect's name. Joining driver's license records to the person table resolved this:

```
CREATE TABLE full_data AS
(SELECT dl.age, dl.height, dl.hair_color, dl.gender, dl.plate_number,
       dl.car_make, dl.car_model, p.name, p.ssn,
       p.address_street_name, p.id
 FROM drivers_license AS dl
 LEFT JOIN person AS p ON dl.id = p.license_id);

SELECT * FROM full_data
WHERE plate_number LIKE 'H42W%'
OR plate_number LIKE '%H42W%'
OR plate_number LIKE '%H42W';
```

The intersection of the gym-membership shortlist and the license-plate shortlist pointed to exactly one name: Jeremy Bowers — the shooter.

6. Post-Analysis and Insights

With the shooter identified, the investigation pulled his interview record and uncovered that the crime was commissioned, not personal:

```
SELECT * FROM interview WHERE person_id = 67318;
```

Jeremy Bowers' statement supplied five independent descriptive filters for the true mastermind: an approximate height range (65"–67"), red hair, female gender, a Tesla Model S, and three attendances at the SQL Symphony Concert in December 2017. Applying all five as compound WHERE conditions against the full_data view isolated the financial backer of the murder:

```
SELECT * FROM full_data
WHERE height BETWEEN 65 AND 67
AND hair_color = 'red'
AND gender = 'female'
AND car_make = 'Tesla'
AND car_model = 'Model S';
```

Key insights from the full workflow:

- Redundant, partial clues (a membership prefix, a plate fragment) are far more powerful when combined than any single clue alone — this is the practical value of multi-column, multi-table filtering.
- Materializing intermediate tables (suspects, full_data) made the investigation auditable and repeatable — any later reviewer can re-run the same queries and reach the same conclusion.
- The case had two distinct culprits at two distinct analytical layers: an executor (Jeremy Bowers) found through physical evidence, and a mastermind found through behavioral/demographic filtering on the same underlying person and drivers_license tables.
- LEFT JOIN was essential once records needed to be resolved across tables with different keys (license_id vs. id) — a common real-world pattern when identity data is scattered across systems.

7. Data Visualizations & Charts

Although the underlying analysis is entirely relational, visualizing the shape of the investigation helps communicate the case to a non-technical audience — for example, a courtroom or a case-review board.

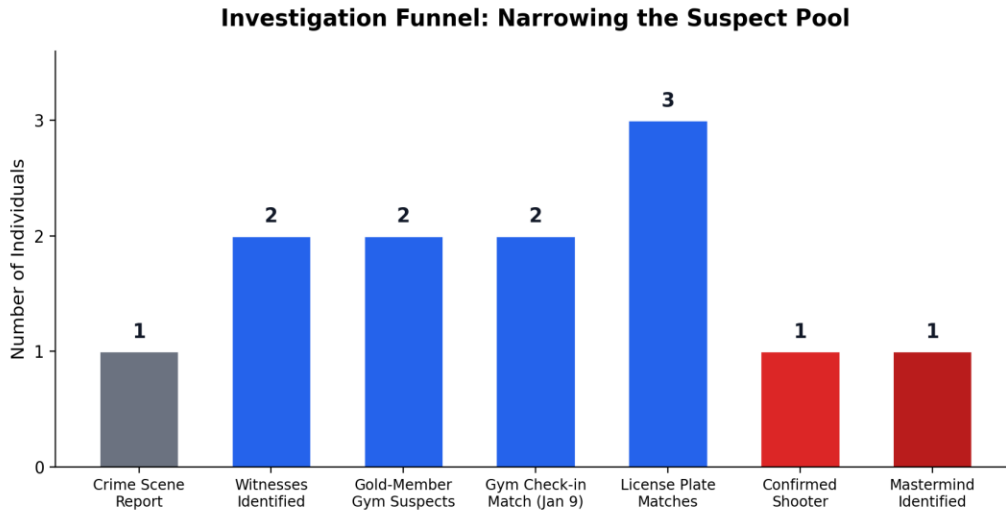


Figure 1. The suspect pool narrowed and re-widened as new independent clues were applied — first two witnesses, then two gym suspects, then three plate matches, collapsing to one shooter and one mastermind.

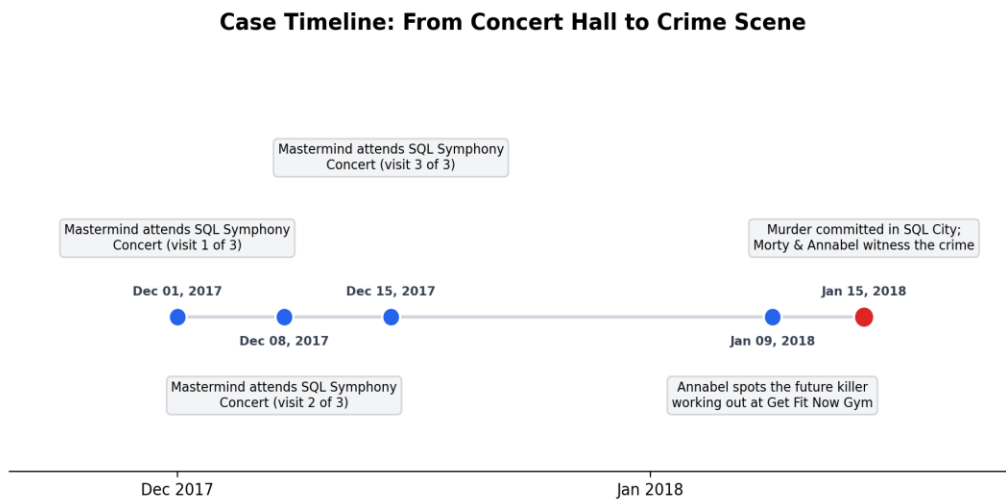


Figure 2. Reconstructed timeline of key events, from the mastermind's concert attendances in December 2017 through the murder on January 15, 2018.

Both figures were generated directly from the dates and counts established during the SQL analysis, illustrating how query results can be translated into investigative exhibits.

8. Recommendations and Observations

8.1 Recommendations

- Standardize partial-match searches: when witnesses provide fragments (e.g., partial plates), always search LIKE '%fragment%' rather than assuming the fragment's position within the field.

- Persist intermediate results as tables (as done with suspects and full_data) for any investigation with more than two filtering stages — this improves auditability and speeds up iterative querying.
- Cross-index behavioral records (gym check-ins, event attendance) with identity and vehicle records earlier in an investigation, since behavioral data often breaks ties that demographic data alone cannot.
- Build reusable views (not just one-off tables) for common joins like person $\bowtie$ drivers_license, since this join was needed twice in this case alone.

8.2 Observations

- The dataset design deliberately introduces red herrings (two gym members checked in on the same day) to test whether an analyst pursues a second corroborating clue instead of stopping at the first apparent match.
- SQL's declarative filtering (WHERE, LIKE, BETWEEN) proved sufficient to solve a multi-suspect case without any external statistical or machine-learning tooling — a reminder that well-modeled relational data can answer surprisingly complex questions.

9. Conclusion

This project demonstrates how a structured, step-by-step SQL investigation — moving from a single crime scene report through witness interviews, membership records, and vehicle data — can resolve a layered criminal case involving both an executor and a mastermind. Jeremy Bowers was identified as the shooter through the intersection of a gym membership clue and a partial license plate, while the woman who hired him was subsequently identified through a compound filter built entirely from his own testimony: physical description, vehicle, and event attendance.

Beyond solving the specific case, the project illustrates transferable SQL and analytical skills: pattern matching with LIKE, sorting to resolve descriptive superlatives ("last house on the street"), CREATE TABLE AS for reusable intermediate results, and LEFT JOIN for reconciling records across mismatched keys. These are the same techniques used in real-world fraud investigation, customer analytics, and operational data reconciliation.

10. References & Appendices

10.1 Reference

Dataset schema and case design adapted from the open-source "SQL Murder Mystery" project (Knight Lab, Northwestern University).

10.2 Appendix A — Case Personnel

Role	Name	Key Identifying Detail
Witness 1	Morty Schapiro	Last house on Northwestern Dr.

Witness 2	Annabel Miller	103 Franklin Ave.
Suspect (cleared)	Joe Germuska	Gold member, ID 48Z7A
Shooter	Jeremy Bowers	Gold member, ID 48Z55; plate incl. H42W
Mastermind	Identified via full_data compound filter	Red hair, 65"–67", Tesla Model S

10.3 Appendix B — Full Annotated SQL Script

The complete, unedited SQL script used throughout this investigation is provided below for reproducibility.

```

SELECT * FROM crime_scene_report
WHERE date = 20180115 AND city = 'SQL City' AND crime_type = 'murder';

SELECT * FROM person
WHERE address_street_name = 'Northwestern Dr'
ORDER BY address_number DESC;  -- Witness 1: Morty Schapiro

SELECT * FROM person
WHERE name LIKE 'Annabel%' AND address_street_name = 'Franklin Ave';
-- Witness 2: Annabel Miller

SELECT * FROM interview WHERE person_id IN (14887, 16371);

CREATE TABLE suspects AS
(SELECT * FROM get_fit_now_member
 WHERE id LIKE '48%' AND membership_status = 'gold');
SELECT * FROM suspects;

SELECT * FROM get_fit_now_check_in
WHERE check_in_date = 20170109
AND membership_id IN ('4D5R1', '4KB72');

SELECT * FROM drivers_license
WHERE plate_number LIKE 'H42W%'
OR plate_number LIKE '%H42W%'
OR plate_number LIKE '%H42W';

CREATE TABLE full_data AS
(SELECT dl.age, dl.height, dl.hair_color, dl.gender, dl.plate_number,
      dl.car_make, dl.car_model, p.name, p.ssn,
      p.address_street_name, p.id
 FROM drivers_license AS dl
 LEFT JOIN person AS p ON dl.id = p.license_id);

SELECT * FROM full_data
WHERE plate_number LIKE 'H42W%'
OR plate_number LIKE '%H42W%'
OR plate_number LIKE '%H42W';
-- Shooter identified: Jeremy Bowers

SELECT * FROM interview WHERE person_id = 67318;

SELECT * FROM full_data

```

```
WHERE height BETWEEN 65 AND 67  
AND hair_color = 'red'  
AND gender = 'female'  
AND car_make = 'Tesla'  
AND car_model = 'Model S';  
-- Mastermind identified
```